

PSET 4 Making Decisions

Your name

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Assignment Instructions:

1. Read the scenario.
 2. Make a decision tree for this scenario in response to Question 1.
 3. Respond to the remaining questions.
 4. When you complete this document, upload the file on the Canvas Assignment page.
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Part I: Scenario

Imagine that you've been building up savings so that you can have an "emergency fund." Since you only plan to use this money to deal with unforeseen circumstances, you don't expect to need any of it in the next year. As a result, you're considering different options for how you might invest it so that it can grow, giving you a potentially larger cushion for the future. Let's assume you're looking to invest \$5,000 and are considering three options.

The economy is currently expanding so the first option you are considering is investing in a stock market index fund. This fund tracks the performance of the stock market overall rather than any particular company or group of companies. You estimate that there is a 15% chance that the economy enters a recession next year, in which case you expect to lose 10% of your investment. However, if the economy continues expanding, you expect to earn a 10% return.

The second option you are considering is to put money into a mutual fund of bonds. This is a combination of bonds from established corporations and the U.S. government. If the economy continues expanding, you expect to earn a 4% return from this investment. In the event that the economy does enter a recession, you think that bonds will seem like a safer investment and become more attractive, which would increase the return on your bond investment slightly to 5%.

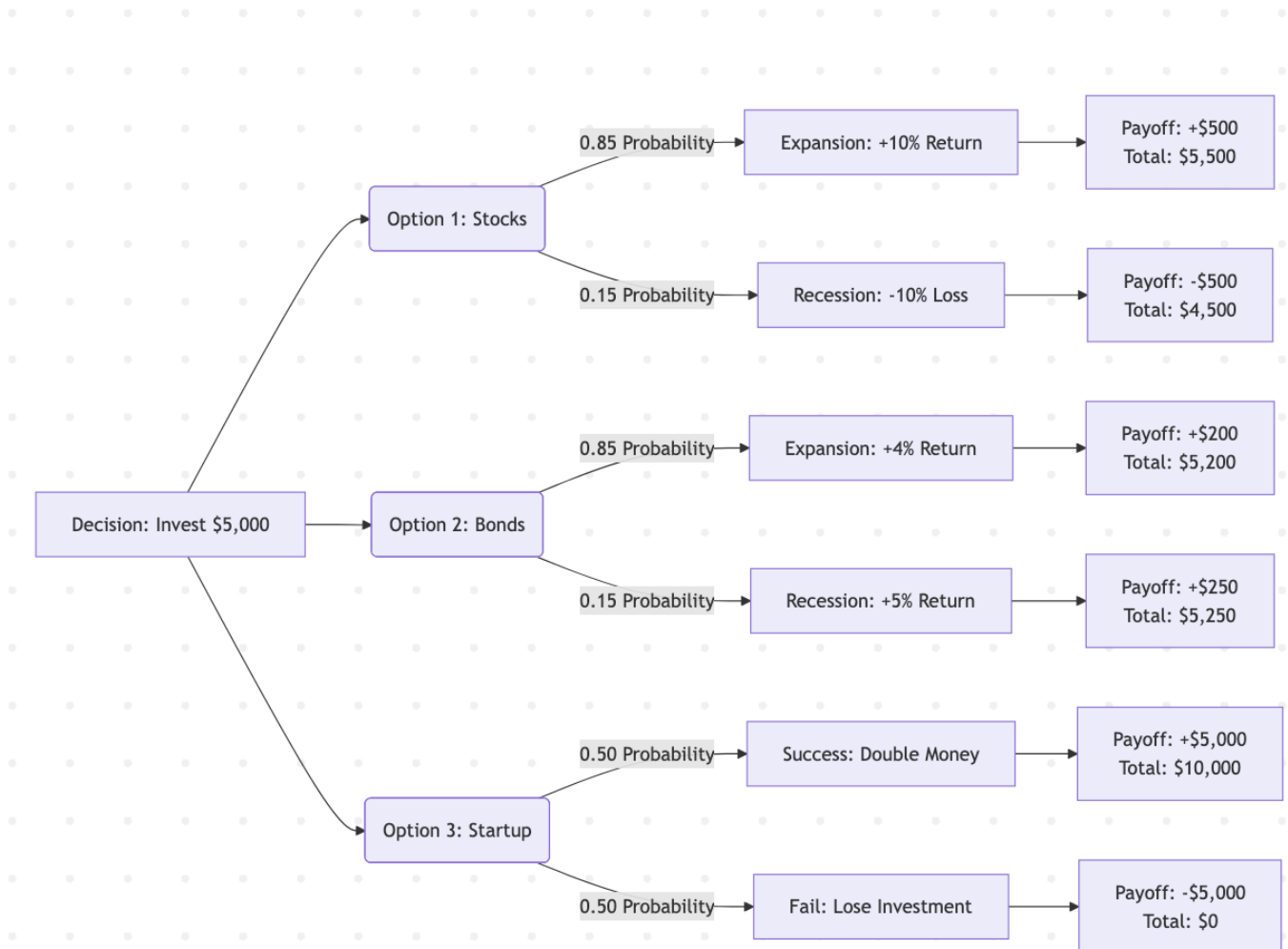
The third option you are considering is funding a friend's startup business. Your friend invited you to invest because if the business hits its growth targets for the next year, it will be able to pay you back double what you put in. However, you know that startups are risky, so you think that the company is just as likely to fail in the next year as it is to succeed. If it fails, you would lose your investment. (Assume the odds that the startup fails are unrelated to whether or not the economy enters a recession.)

Part II: Questions

Question 1: Use any method you prefer to make a decision tree and include it in this assignment below, including:

- Sketch by hand, take a picture and insert the image
- Modify the decision tree template provided during the live lecture class
- Alternatively, use another method to make a decision tree

For this first question, you only have to construct the tree (not solve it). Make sure that your tree includes choices, probabilities, and payoffs.



Graph made with mermaid code syntax ^

Question 2:

Solve the decision tree by calculating the expected value of each choice. Using these calculations alone, what does this tree suggest about where to invest your \$5,000? Show your work in calculating the expected values. Write your answer to the question "What does this tree suggest about where to invest your \$5,000?" in the same text box below.

To find the best option, I'm calculating the EV for each scenario. This just involves multiplying the probability of each outcome by its financial payoff and summing them up to see the "average" result over time.

Here is the breakdown:

Option 1: Stock Market Index Fund

- Scenario A (Expansion): There is an 85% chance of earning \$500 (10% of \$5,000).

*Calculation: $0.85 * \$500 = \425*

- Scenario B (Recession): There is a 15% chance of losing \$500 (10% of \$5,000).

*Calculation: $0.15 * -\$500 = -\75*

- Net Expected Value: $\$425 + (-\$75) = \$350$

Option 2: Bond Mutual Fund

- Scenario A (Expansion): There is an 85% chance of earning \$200 (4% of \$5,000).

*Calculation: $0.85 * \$200 = \170*

- Scenario B (Recession): There is a 15% chance of earning \$250 (5% of \$5,000).

*Calculation: $0.15 * \$250 = \37.50*

- Net Expected Value: $\$170 + \$37.50 = \$207.50$

Option 3: Startup Business

- Scenario A (Success): There is a 50% chance of doubling the money (profit of \$5,000).

*Calculation: $0.50 * \$5,000 = \$2,500$*

- Scenario B (Failure): There is a 50% chance of losing the entire investment (-\$5,000).

*Calculation: $0.50 * -\$5,000 = -\$2,500$*

- Net Expected Value: $\$2,500 - \$2,500 = \$0$

Answer:

Based strictly on these calculations, the decision tree suggests that Option 1 (Stocks) is the best investment. It offers the highest expected value (\$350) compared to Bonds (\$207.50) and the Startup (\$0).

Question 3:

Which option would you choose, and why? Does your choice deviate from what the decision tree suggests is the optimal choice in Question 2? Why or why not? (100-200 words)

I would choose Option 1 (the Stock Market Index Fund).

My reasoning is largely driven by the Expected Value calculation in Question 2, where stocks clearly outperformed the other options with a projected return of \$350. While I acknowledge the 15% risk of a recession, the prompt notes that I don't expect to need this money in the upcoming year. This gives me a slightly higher risk tolerance; I can afford to ride out a temporary 10% dip in exchange for the high probability (85%) of getting a significant 10% return.

The startup is simply too volatile for an emergency fund (effectively a coin toss), and the bonds, while safe, leave too much potential profit on the table given the high likelihood of economic expansion.

Question 4:

How would your choice change if your risk tolerance were lower? Consider the impact of potential losses on your overall financial goals and peace of mind. Which option best aligns with a conservative approach to preserving your emergency fund?? (100-200 words)

If my risk tolerance were lower, my choice would definitely shift to Option 2 (the Bond Mutual Fund).

This option aligns best with a conservative approach because it offers "capital preservation." In Option 1, I risk losing \$500 if the economy tanks. In Option 3, I risk losing everything. Option 2 is the only scenario where I make a profit regardless of what the economy does (\$200 if it expands, \$250 if it shrinks).

For an emergency fund, peace of mind is often more valuable than maximizing returns. Knowing that my \$5,000 principal is safe and will grow slightly no matter the economic climate makes this the responsible choice for a risk-averse investor.

Question 5:

Assume each investment option incurs an annual fee (e.g., fund management or account fees) of 1%. How would these fees affect the net expected value? Show your work.

First, I need to determine the cost of the fee. A 1% fee on a \$5,000 investment equals \$50. I will subtract this fixed cost from the Net Expected Values I calculated in Question 2 to find the new "Net EV."

Calculations:

- Option 1 (Stocks): Previous EV (\$350) - Fee (\$50) = \$300*
- Option 2 (Bonds): Previous EV (\$207.50) - Fee (\$50) = \$157.50*
- Option 3 (Startup): Previous EV (\$0) - Fee (\$50) = -\$50*

Interpretation:

The fees reduce the overall profit for every option, but they don't actually change my final decision.

- 1. Stocks are still the winner with an EV of \$300.*
- 2. Bonds are still second with an EV of \$157.50.*
- 3. The Startup drops to last place with a negative EV of -\$50.*

Essentially, the fee makes the startup an even worse proposition (statistically losing money), but the trade-off between Stocks and Bonds remains the same as it was before the fees were applied.

Question 6:

Would you consider diversifying by splitting your \$5,000 across multiple options? If so, what portion would you allocate to each, and why? (100-200 words)

Yes, I would definitely consider diversifying. I would propose splitting the \$5,000 by putting 70% (\$3,500) into Option 1 (Stocks) and 30% (\$1,500) into Option 2 (Bonds), while avoiding the startup entirely.

The reasoning here is to create a hedge against the recession scenario. If the economy expands, the bulk of my money (the stock portion) captures that high 10% growth. However, if the economy hits a recession, the stocks will lose value, but the bond portion will actually see its interest rate rise to 5%. This bond profit helps offset the stock losses, smoothing out the volatility. Since this is an emergency fund, this balanced approach reduces the risk of a sharp drawdown while still offering better growth potential than going 100% into bonds.

Grading Rubric:

(100 points max)

Question 1: Construct the Decision Tree (10 points)

- **Structure (4 points):** Make sure your decision tree clearly shows all three investment options as branches, with further branches showing different economic outcomes.
- **Probabilities (3 points):** Each branch should include the correct probability as given in the scenario.
- **Payoffs (3 points):** Each outcome should display the correct financial payoff based on the expected return.

Question 2: Solve the Decision Tree and Calculate Expected Values (25 points)

- **Expected Value Calculations (15 points):** Calculate the expected values (EV) accurately by multiplying each outcome's probability by its payoff and then adding up these values for each investment option.
- **Show Your Work (5 points):** Show your calculations step-by-step for each option.
- **Interpretation (5 points):** Explain which investment option the EV suggests is the best choice.

Question 3: Decision and Rationale (15 points)

- **Clear Decision (5 points):** Clearly state which investment you would personally choose.
- **Reasoning (10 points):** Explain whether your choice aligns with the EV calculation and discuss any personal factors (like your risk tolerance) that influenced your decision.

Question 4: Impact of Lower Risk Tolerance (15 points)

- **Reflection on Risk (10 points):** Explain how having a lower risk tolerance would affect your investment decision, especially in terms of financial goals and peace of mind.
- **Conservative Choice (5 points):** Identify which option would be the safest or most conservative choice, and explain why.

Question 5: Effect of Fees on Expected Value (20 points)

- **Fee Calculation (10 points):** Apply a 1% fee to each option and calculate the new expected value (net EV).
- **Show Work (5 points):** Display your calculations step-by-step to show how the fees impact the EV.
- **Interpretation (5 points):** Explain how the fees affect your decision and whether it changes which option you prefer.

Question 6: Consideration of Diversification (15 points)

- **Propose an Allocation (10 points):** If you choose to diversify, explain how you would split the \$5,000 across the different options and why.
- **Reasoning (5 points):** Discuss why or why not you think diversification would be helpful, considering risk reduction and potential returns.